NS4150B Audio Amplifier Datasheet Translation

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User prompt: can you convert this pdf to English? I can't read Japanese.

Response: Here is the complete English translation of the **NS4150B 3.0W Monophonic Class-D Audio Power Amplifier** datasheet:

Shenzhen Nsiway Technology Co., Ltd.

NS4150B 3.0W Monophonic Class-D Audio Power Amplifier

1. Features

- Operating Voltage Range: 3.0V ~ 5.0V
- Output Power: 2.8W ($5\text{V}/4\,\Omega$, $\text{THD} = 10\%$); $0.1\%\ \text{THD}$ ($0.5\text{W}/3.6\text{V}$)
- Up to 88% Efficiency
- High PSRR: -80dB (217Hz)
- Filterless Class-D Architecture
- Excellent Full-Bandwidth EMI Suppression
- Excellent "Power-Up / Power-Down" Noise Suppression
- Low Quiescent Current: 4mA (3.6V supply, No load)
- Over-Current Protection, Thermal Protection, Under-Voltage Lockout (UVLO)
- MSOP8 Package

2. Applications

- Tablet Computers
- Dash Cams
- Bluetooth Speakers

3. Description

The NS4150B is an ultra-low EMI, filterless, 3W monophonic Class-D audio power amplifier. Utilizing advanced technology, the NS4150B drastically reduces EMI interference across the entire

bandwidth, minimizing impact on surrounding components.

The NS4150B features built-in over-current protection, thermal protection, and under-voltage protection, effectively safeguarding the chip from damage under abnormal operating conditions. By leveraging spread-spectrum technology and fully optimizing a new circuit design, efficiencies up to 90% make it ideally suited for portable audio products.

The NS4150B's filterless PWM modulation architecture and internal gain configuration reduce external component counts, PCB footprint, and system cost. Available in an MSOP-8 package, its rated operating temperature range is $-40^{\circ}\mathrm{C}$ to $85^{\circ}\mathrm{C}$.

5. Pin Configuration

Pin Number	Pin Name	Pin Description
1	CTRL	Operating mode control; Low level = Shutdown
2	Bypass	Internal common-mode voltage bypass capacitor pin; Connect $1\mu\mathrm{F}$ capacitor to GND
3	INP	Positive audio input
4	INN	Negative audio input
5	VoN	Negative audio output
6	VCC	Power supply input and audio power stage supply pin
7	GND	Ground
8	VoP	Positive audio output

6. Absolute Maximum Ratings

Parameter	Minimum	Maximum	Unit	Description
Supply Voltage (V_{DD})	-0.3	5.25	V	
INP / INN / CTRL Pins	-0.3	$V_{DD}+0.3$	V	
Storage Temperature	-65	150	$^{\circ}\mathrm{C}$	
ESD Withstand Voltage	—	± 4000	V	
Junction Temperature	—	150	$^{\circ}\mathrm{C}$	

Parameter	Minimum	Maximum	Unit	Description
Recommended Operating Temperature	-40	85	$^\circ\mathrm{C}$	
Recommended Operating Voltage	2.2	5.0	V	
Thermal Resistance (MSOP-8)	—	190	$^\circ\mathrm{C/W}$	
Latch-up	—	± 150	mA	
Soldering Temperature	—	220	$^\circ\mathrm{C}$	Within 15 seconds

Note: Exceeding the absolute maximum ratings may cause permanent chip damage. Extended exposure to these limits may affect reliability and lifespan.

7. Electrical Characteristics

Test Conditions (unless otherwise specified): $T = 25^\circ\mathrm{C}$, $V_{DD} = 4.8\mathrm{V}$.

Symbol	Parameter	Test Conditions	Min	Typ	Max	Unit
V_{DD}	Supply Voltage	—	3.0	4.8	5.0	V
I_{DD}	Quiescent Supply Current	$V_{IN} = 0\mathrm{V}$, $V_{DD} = 3.6\mathrm{V}$, No Load	—	3.0	—	mA
I_{SD}	Shutdown Leakage Current	$V_{DD} = 3.6\mathrm{V}$, $\mathrm{CTRL} = 0\mathrm{V}$	—	0.1	10	$\mu\mathrm{A}$
V_{OS}	Output Offset Voltage	$V_{IN} = 0\mathrm{V}$, $V_{DD} = 3.0\mathrm{V}$ to $5.25\mathrm{V}$	—	5	20	mV
f_{SW}	Switching Frequency	$V_{DD} = 3.0\mathrm{V}$ to $5.25\mathrm{V}$	—	400	—	kHz
P_O	Output Power	$\mathrm{THD} = 1\%$, $f = 1\mathrm{kHz}$, $V_{DD} = 5\mathrm{V}$, $R_L = 4\,\Omega$ $\mathrm{THD} = 1\%$, $f = 1\mathrm{kHz}$, $V_{DD} = 5\mathrm{V}$, $R_L = 8\,\Omega$ $\mathrm{THD} = 10\%$, $f = 1\mathrm{kHz}$,	—	2.0 1.3 2.8	—	W

Symbol	Parameter	Test Conditions	Min	Typ	Max	Unit
		$V_{DD} = 5\text{V}$, $R_L = 4\,\Omega$				
		$\text{THD} = 10\%$, $f = 1\text{kHz}$, $V_{DD} = 5\text{V}$, $R_L = 8\,\Omega$		1.7		
$\text{THD} + \text{N}$	Total Harmonic Distortion + Noise	$V_{DD} = 3.6\text{V}$, $f = 1\text{kHz}$, $R_L = 8\,\Omega$, $P_O = 0.1\text{W}$	—	0.15	—	%
		$V_{DD} = 3.6\text{V}$, $f = 1\text{kHz}$, $R_L = 4\,\Omega$, $P_O = 0.5\text{W}$		0.10		
PSRR	Power Supply Rejection Ratio	217Hz	—	−80	—	dB
		20kHz		−72		
CMRR	Common Mode Rejection Ratio	—	—	−70	—	dB
η	Efficiency	$P_O = 0.6\text{W}$, $f = 1\text{kHz}$, $V_{DD} = 3.6\text{V}$, $R_L = 8\,\Omega$	—	90	—	%
V_{IH}	CTRL Input High Level	—	1.2	—	V_{DD}	V
V_{IL}	CTRL Input Low Level	—	0	—	0.2	V
T_{st}	Start-up Time	—	—	120	—	ms
T_{sd}	Shutdown Time	—	—	80	—	ms

9. Application Information

9.1 Basic Functional Description

The NS4150B is an ultra-low EMI, filterless, 3W monophonic Class-D audio power amplifier. Under a 5V supply, it can deliver 3W of power into a $4\,\Omega$ load at up to 90% efficiency.

The chip incorporates over-current protection, thermal protection, and under-voltage lockout to

safely shut down the device under abnormal conditions and automatically resume operation once conditions normalize.

9.2 Filterless Operation

The NS4150B uses a filterless PWM modulation scheme, eliminating traditional LC filters required by Class-D amplifiers to maximize efficiency and achieve a smaller footprint and lower system cost.

9.3 Power-Up and Power-Down Noise Suppression

Built-in transient suppression circuitry effectively eliminates pop and click noises during power-up, power-down, wake-up, and shutdown sequences.

9.4 EMI Enhancement Technology

Advanced built-in EMI reduction techniques significantly suppress high-frequency interference across the full bandwidth, minimizing emissions affecting nearby components (compliant with FCC Part 15 Class B standards).

9.5 CTRL Pin Settings

CTRL Pin Level	Operating Mode
H	Active / Working
L	Shutdown

9.7 Protection Circuitry

- **Over-Current Protection (OCP):** Shuts down the output stage during short circuits and auto-recovers when cleared.
- **Thermal Protection (OTP):** Shuts down the device if internal temperature exceeds safe limits, resuming operation upon cooling.
- **Under-Voltage Lockout (UVLO):** Powers down the chip if supply voltage drops below thresholds, restarting once voltage recovers.

9.8 Power Supply Decoupling Capacitor

Place a $1\mu\mathrm{F}$ ceramic capacitor (low ESR) as close to the V_{DD} and GND pins as possible. For enhanced low-frequency filtering, a $10\mu\mathrm{F}$ or larger bulk capacitor can be added.

9.9 Gain Configuration and Input Resistor

The internal feedback resistor is fixed at $240\,\mathrm{k}\Omega$. The closed-loop voltage gain is given by:

$$A_{VD} = \frac{240\,\text{k}\Omega}{R_{in}}$$

(where R_{in} is the external input resistor).

9.10 Input Filter

Input DC-blocking capacitors (C_{in}) and input resistors (R_{in}) form a high-pass filter with a cutoff frequency of:

$$f_c = \frac{1}{2\pi \cdot R_{in} \cdot C_{in}}$$

Smaller coupling capacitors help reduce switch/pop noise without impacting performance, as typical miniature speakers cannot reproduce frequencies below $100\text{Hz} - 150\text{Hz}$.

9.11 Ferrite Beads and Capacitors

If long trace routing or proximity to EMI-sensitive circuits requires extra filtering, ferrite beads and 1nF capacitors to ground should be placed close to the V_OP and V_ON output pins.

10. Package Information (MSOP-8 Dimensions)

Symbol	Min (mm)	Nom (mm)	Max (mm)
A	—	—	1.10
A1	0.05	—	0.15
A2	0.75	0.85	0.95
A3	0.30	0.35	0.40
b	0.22	—	0.38
b1	0.28	0.30	0.33
c	0.15	—	0.20
c1	0.14	0.152	0.16
D	2.90	3.00	3.10
E	4.70	4.90	5.10

Symbol	Min (mm)	Nom (mm)	Max (mm)
E1	2.90	3.00	3.10
e	—	0.65 BSC	—
L	0.40	—	0.80
L1	—	0.95 BSC	—
θ	0°	—	8°
